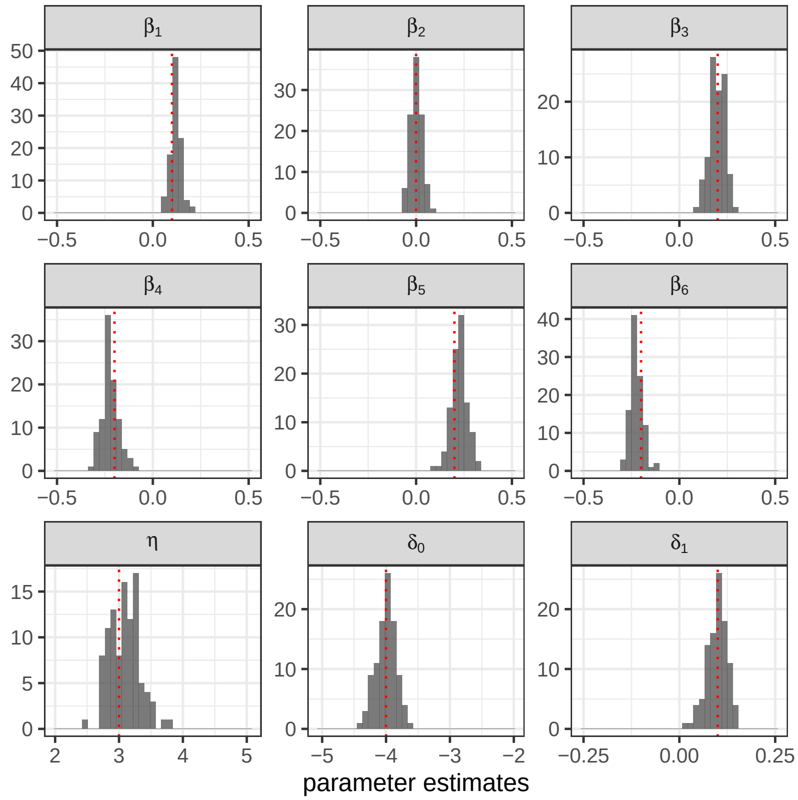

Notes: Each plot corresponds to one parameter in the search model. The red dotted line shows the true parameter value. The histogram shows the parameter estimates across 100 Monte Carlo datasets. The range of the horizontal axis equals the parameter's range in Θ.

Figure 4: NNE's Estimates for Search Model in Monte Carlo

Some biases may be noticeable but are relatively small.

Figure 5 plots the estimates by SMLE under three smoothing factors: $\lambda = 3$, 7, and 10. In particular, we include $\lambda = 7$ because it is the optimal smoothing factor for SMLE here. Specifically, we use a grid of λ from 1 to 15. For each point on the grid, we repeat SMLE over 100 Monte Carlo datasets to compute a RMSE. The RMSE is smallest at $\lambda = 7$. We give the details in Appendix A.3. It is worth noting that obtaining this optimal λ not only is computationally expensive but also requires knowing the true value of $\boldsymbol{\theta}$ (to calculate RMSE). More generally, it is known that finding optimal λ is difficult in the estimation of search models.

Figure 5 shows that the estimates by SMLE are sensitive to λ. We also see that all three values of λ lead to some biases. As we will show in a moment, both model fit and counterfactual analysis are more accurate with NNE than SMLE, even when SMLE uses the optimal $\lambda = 7$. It is also worth noting that the SMLE here has a higher computational cost than the NNE in Figure 4. We now turn to a more detailed comparison of computational costs.

Figure 6 plots the estimation accuracy of SMLE and NNE as we vary computation costs (by varying L^* and R). We measure estimation accuracy by RMSE. We measure computational costs in two ways: simulation burden and compute time. The left plot of Figure 6 shows the simulation